OOF® Global Interest Heat Map™

Technical, data, usability and discovery audit

82 / 100	OVERALL RESULT
GOOD	Production-ready with targeted transparency and resilience improvements recommended.

Executive summary

PASS	Interactive map	Rendered correctly on desktop and mobile; selector, zoom, reset and country focus worked without console errors.
PASS	Data freshness	Live dataset generated 30 Sep 2026 at 09:40 UTC from a rolling 14-day Cloudflare analytics window.
PASS	Automation	Latest two scheduled GitHub Actions runs succeeded. Nine of the latest ten runs succeeded; one transient build failure recovered the next day.
PASS	Privacy	Public JSON exposes only ISO country code, classification and optional momentum. Visitor counts, IP data and device identifiers are explicitly rejected.
IMPROVE	Transparency	The page does not visibly state the 14-day window, generation timestamp, analytics source, algorithm version or freshness status.
IMPROVE	SEO	The generated meta description includes loading text and control symbols, weakening search-result quality and AI summaries.

Audit scope

Reviewed the live page, public JSON payload, map JavaScript, responsive CSS, data-generation script, validator, unit tests, scheduled GitHub workflow and the latest ten workflow runs. The audit did not access raw Cloudflare visitor records or repository secrets.

1. Architecture and data pipeline

The implementation uses a static HTML page, locally hosted Leaflet 1.9.4, Natural Earth 1:50m country geometry and a privacy-reduced JSON dataset generated by GitHub Actions.

1	**Cloudflare Web Analytics**	Aggregated page-load visits grouped by country and date; bots excluded; host restricted to originopenfoundation.org.
2	**14-day processing**	Rolling 14-day window, minimum 10 visits and 2 active days before a country can qualify for classification.
3	**Normalization**	Visits per million residents adjusted by reliability based on visit count and active days; World Bank population data used.
4	**Public reduction**	Only ISO code, status and optional momentum are published. Internal metrics are retained as a 90-day workflow artifact.
5	**Daily publication**	Scheduled workflow validates and commits the dataset atomically to main1; the last valid dataset is preserved if credentials are unavailable.

Current public state

Generated	30 Sep 2026, 09:40:24 UTC	**HTTP**	200 OK
Cache	max-age=600	**Classified**	3 countries
High	Singapore	**Moderate**	Slovakia
Emerging	United States	**Others**	Insufficient Signal

Assessment

The aggregation and public-reduction model is privacy-conscious and avoids exposing raw audience counts.

Atomic validation before replacement is a strong safeguard against publishing malformed data.

The quantile-based labels are relative. With only three qualifying countries, a small traffic change can shift labels even when absolute interest is stable.

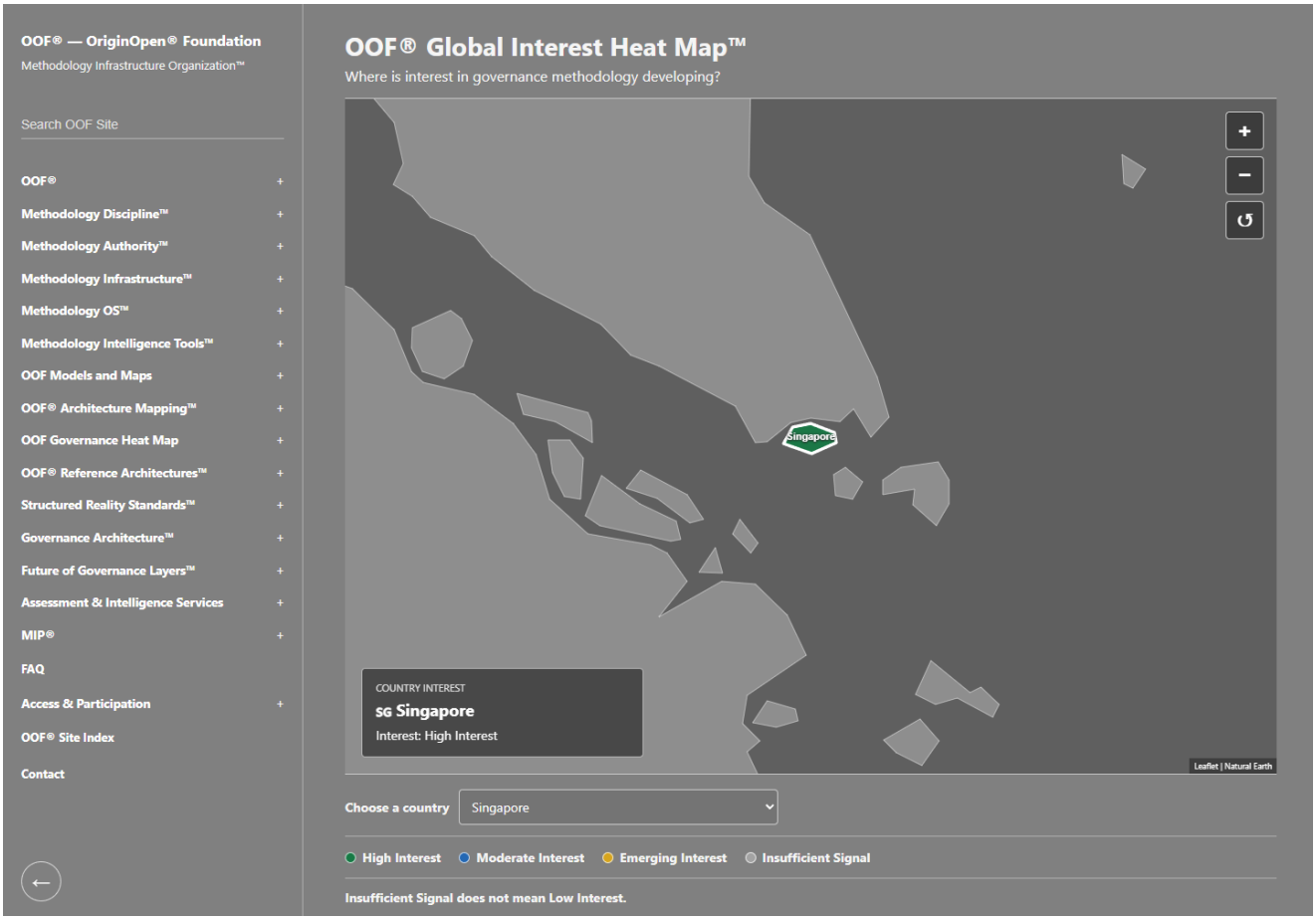
2. Findings and priorities

F-01	**P2**	Freshness and 14-day window are not visible on the page.	Visitors cannot verify when the map was updated or what period the classifications represent.	Show 'Last updated', 'Data window: previous 14 days' and source near the legend.
F-02	**P2**	SEO description contains loading text and control symbols.	Search snippets and AI summaries may look unfinished or describe controls instead of the map's purpose.	Use a curated description that states country-level interest, 14-day aggregation and privacy-safe classification.
F-03	**P2**	Public payload omits algorithm and window metadata.	External auditors cannot reproduce or correctly contextualize a historical classification from public data alone.	Add schemaVersion, algorithmVersion, windowStart and windowEnd; continue excluding counts and personal data.
F-04	**P2**	Automated tests cover classification basics but not the interactive map.	Regressions in rendering, country selection, small-state labels and mobile controls could reach production.	Add browser tests for load, colors, selector, zoom/reset, Singapore callout, no horizontal overflow and fallback mode.
F-05	**P3**	One of the latest ten scheduled runs failed at dataset generation.	The last valid data remained available and later runs recovered, but failure visibility depends on GitHub Actions.	Add failure notification and a stale-data warning when generatedAt exceeds 36 hours.
F-06	**P3**	generatedAt changes every run, causing daily commits even when classifications do not.	Creates repository and deployment churn without a visible information change.	Separate lastCheckedAt from classification revision or commit only when country states change.
F-07	**P3**	The map uses role=application and touch-action:none.	Advanced screen readers and one-finger mobile scrolling may require extra effort inside the large map area.	Retain the selector, document keyboard behavior, and test with NVDA/VoiceOver and real touch devices.

Priority legend: P1 = critical, P2 = important, P3 = improvement. No P1 issue was found.

3. Live desktop verification

The live page was loaded at 1440 x 1000 pixels. Singapore was selected through the native country selector, which focused the map and displayed the correct classification.



Verified behavior

PASS	Map rendered with detailed country geometry and no blank canvas.
PASS	Singapore small-state geometry and English callout remained visible after focus.
PASS	Country card reported 'Singapore - High Interest'.
PASS	Zoom in, zoom out and reset controls were present with accessible labels.
PASS	No JavaScript console or page errors were observed.

4. Mobile, accessibility and performance



Mobile verification

- Tested at 390 x 844 pixels with no horizontal overflow.
- Map viewport remained 338 px wide and approximately 439 px tall after the country card was included.
- Controls measured 46 x 46 px, meeting common touch-target guidance.
- The country card moves below the map on mobile, preventing it from covering country geometry.

Accessibility

- A native select exposes all 236 mapped countries plus the prompt option.
- Map controls have explicit aria-label values and keyboard-focus styling.
- Textual status supplements color, and the page explicitly states that insufficient signal is not low interest.
- A non-interactive fallback and noscript explanation are present.

Performance snapshot

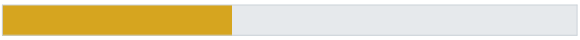
- 12 observed resources; approximately 386 KB transferred and 1.11 MB decoded in the audit browser.
- Interactive state was ready in approximately 1.2-1.6 seconds in the audit environment.

5. Recommended implementation plan

1 - Transparency	Expose freshness, data window, source and methodology link near the legend.	A visitor can identify the exact generation time and rolling 14-day period without opening JSON or source code.
1 - Discovery	Replace generated SEO description and update WebPage JSON-LD with Dataset linkage.	Search description contains no loading state or control symbols; structured data validates.
2 - Reproducibility	Extend public metadata with schemaVersion, algorithmVersion, windowStart and windowEnd.	Public data remains free of visitor counts and identifiers while historical states become interpretable.
2 - Resilience	Add stale-data status and workflow failure notification.	A failed refresh is visible within 24 hours and the page warns when data is older than 36 hours.
3 - Testing	Add automated desktop/mobile browser checks and accessibility smoke tests.	CI verifies rendering, selection, colors, controls, fallback, small countries and horizontal overflow.
3 - Operations	Reduce no-op commit churn and document classification stability constraints.	No deployment occurs when country classifications and momentum are unchanged.

Recommended release decision: keep the current map live. Address Phase 1 items first because they improve trust, search presentation and AI interpretability without changing the classification algorithm.

Scoring breakdown

Functionality	18/20	
Data pipeline	17/20	
Mobile and usability	17/20	
Privacy and security	14/15	
Accessibility	8/10	
SEO and transparency	4/10	
Operations and resilience	4/5	

Appendix A - Verification record

Public data schema	**PASS**	Validator accepted 3 classified countries; no forbidden fields.
Unit tests	**PASS**	2/2 tests passed for ISO resolution, Antarctica exclusion, thresholds and public-data reduction.
Live dataset	**PASS**	HTTP 200, cache max-age=600, generatedAt 2026-09-30T09:40:24Z.
Desktop render	**PASS**	1440 x 1000; selection and focused small-state map verified; no console errors.
Mobile render	**PASS**	390 x 844; no horizontal overflow; mobile country card and controls verified.
Scheduled automation	**PASS WITH NOTE**	Latest two runs succeeded; 9/10 latest runs succeeded; 28 Sep failure occurred in Build next public dataset.
Discovery metadata	**IMPROVE**	Canonical, robots and JSON-LD present; description quality requires correction.

Sources reviewed

Live page: https://originopenfoundation.org/global-interest-heat-map.html

Public dataset: https://originopenfoundation.org/data/oof-global-interest-heat-map.json

Workflow runs: https://github.com/originopenfoundation/OOF-Origin-Open-Foundation-/actions/workflows/global-interest-heat-map.yml

Repository files: global-interest-heat-map.html, assets/js/oof-global-interest-heat-map.js, style.css, tools/update_global_interest_heat_map.py, tools/validate_global_interest_heat_map.py, tests/test_global_interest_heat_map.py and .github/workflows/global-interest-heat-map.yml

Audit limitation

This audit verified the public implementation and automation evidence available on 30 September 2026. It did not inspect raw Cloudflare analytics events, secret values, private workflow artifacts or end-user analytics consent configuration.